

# Images and Graphics – Editing

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*Photo editing software*

# Bitmap graphics - Adobe Photoshop

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Photoshop is one of the most used commercial tools to deal with bitmap graphics

It can also handle some vector elements

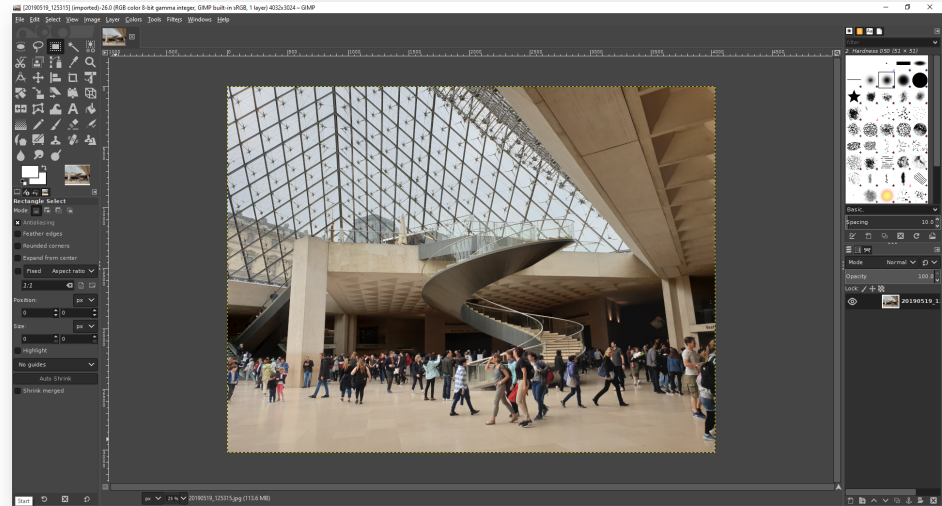


*Photo editing software*

# Bitmap graphics - GIMP

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GIMP (GNU Image  
Manipulation Program) is a  
good (free and open source)  
alternative to Photoshop for  
“standard” purposes



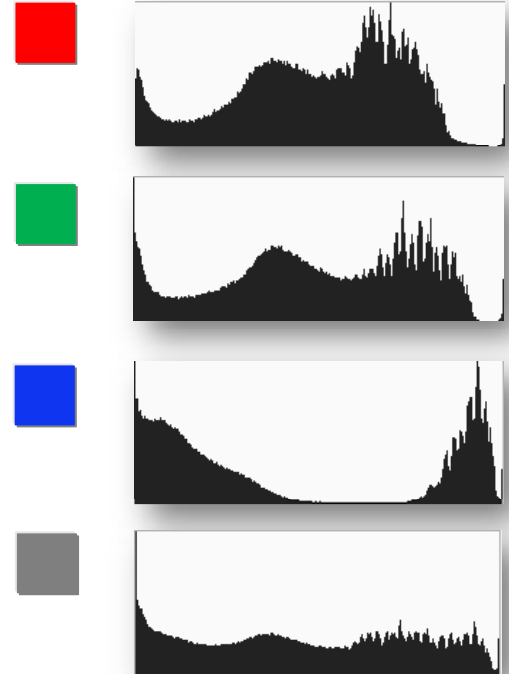
# Image histogram...

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Represents the tonal or luminance distribution in a bitmap image

X axis: 256 levels (0 ÷ 255)

Y axis: number of pixels with a specific tonal or luminance value

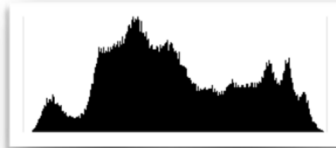
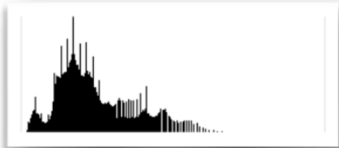




*Photo editing software*

## ... image histogram

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# Global adjustments

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Modify the image (or a part of it) “as a whole”

Some adjustments available in Photoshop, GIMP, and other tools:

- Levels
- Curves
- Color balance
- Hue/lightness/saturation
- Filters

## Example

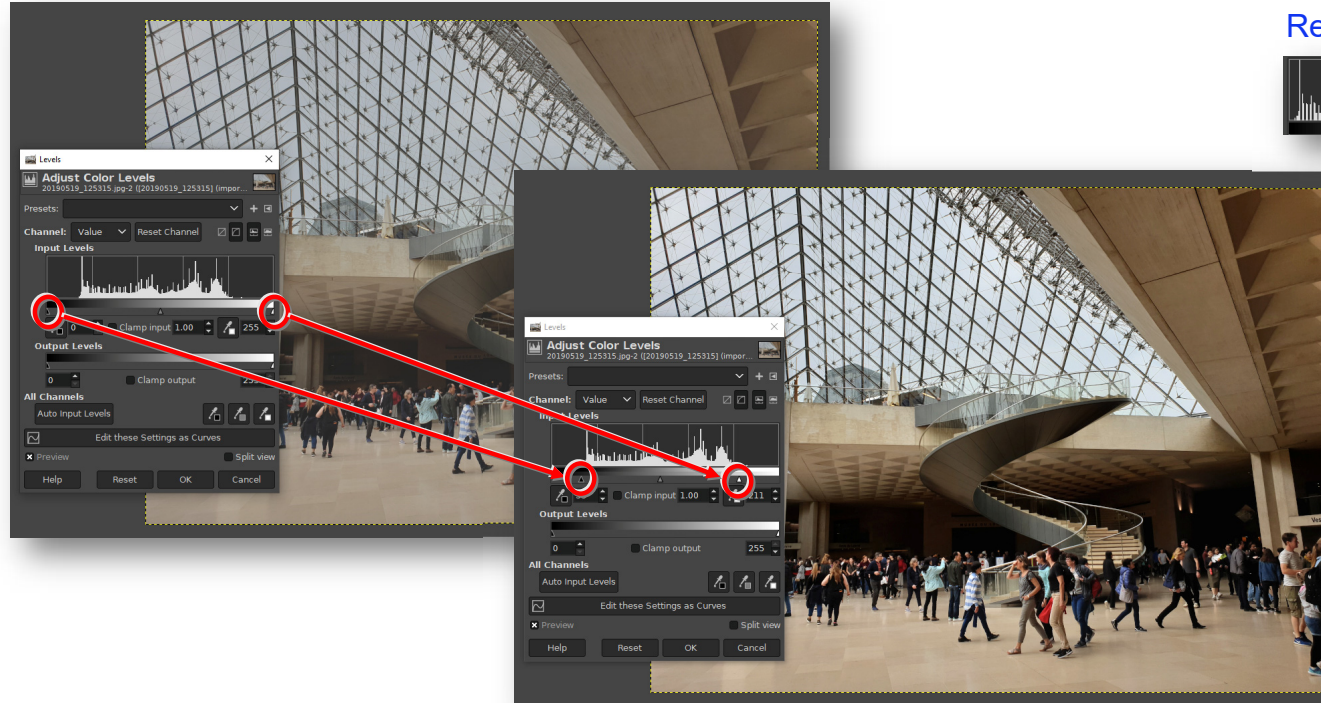
*Original image*



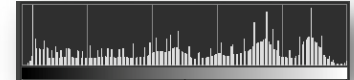
*Image after color balance*



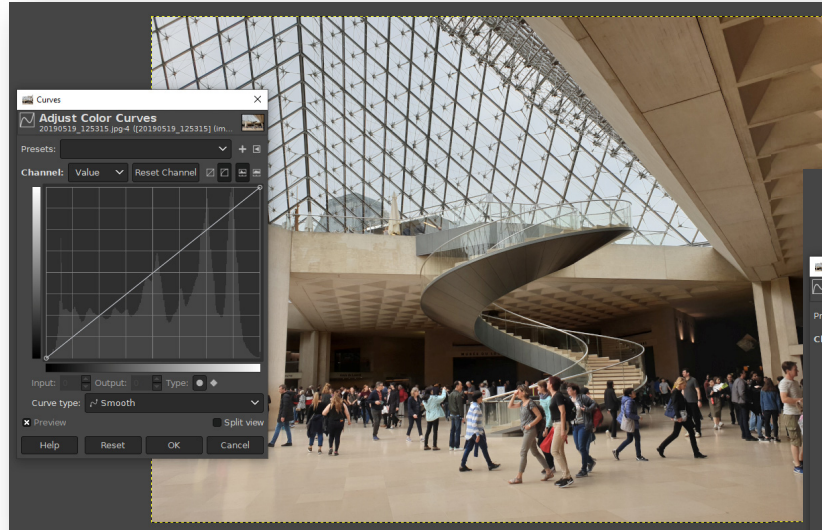
# Levels



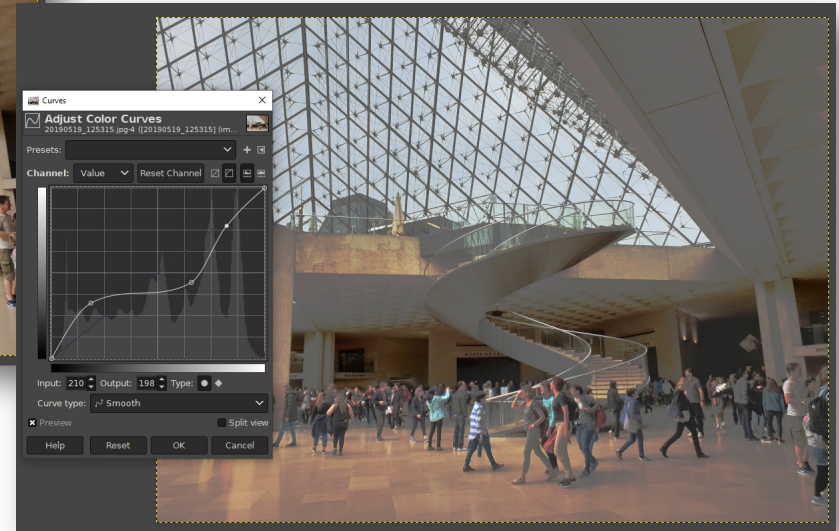
Resulting histogram



# Curves



X-axis → input values  
Y-axis → output values





# Color balance

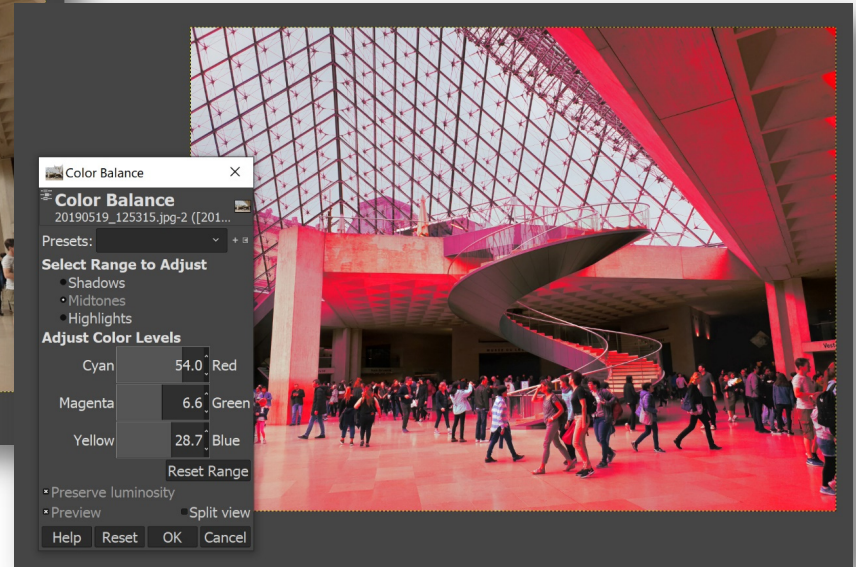
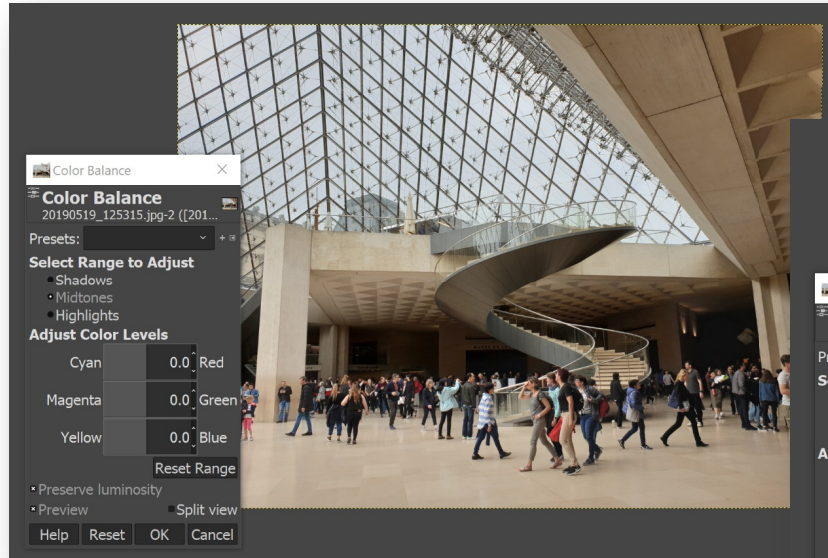
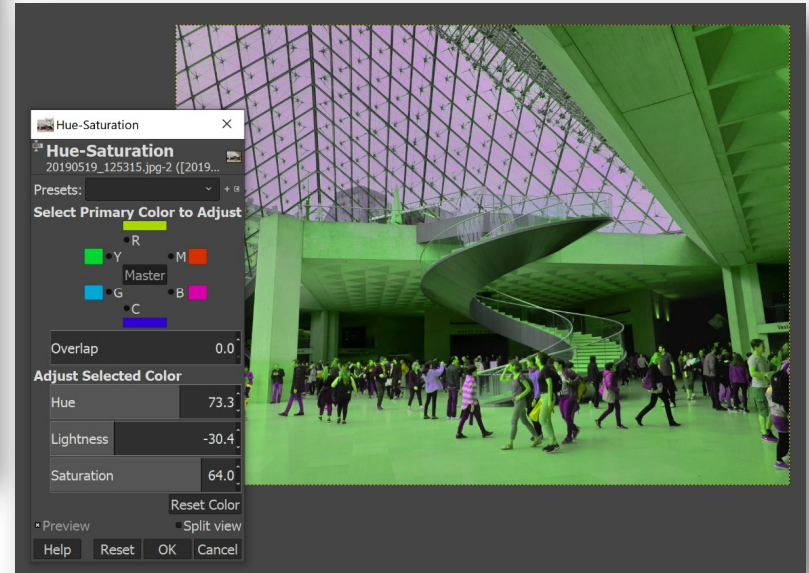
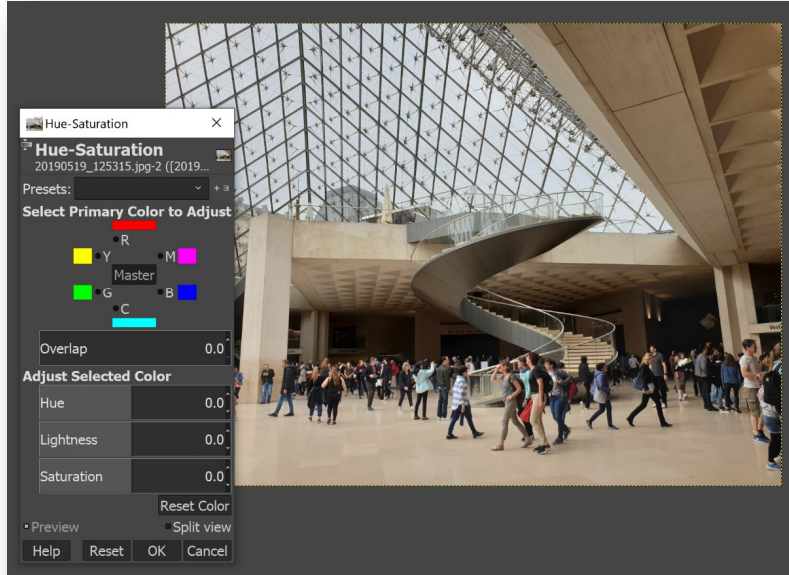


Photo editing software – Typical functions (GIMP)

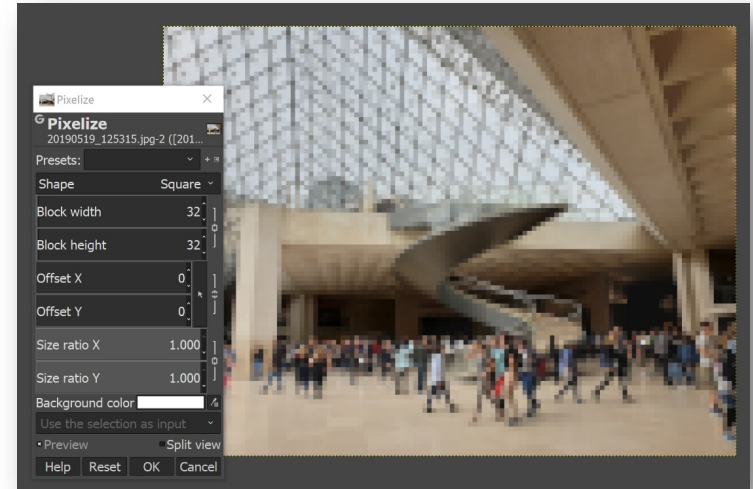
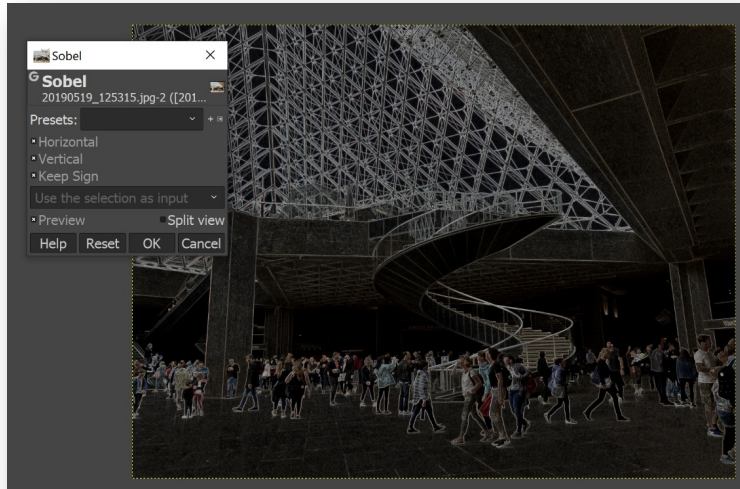
# Hue/lightness/saturation



# Filters

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Can automatically create “special effects” (actually, perform image transformations)



# Local adjustments...

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## Modify circumscribed areas of the image

Functions available (in different forms) in most photo editing software tools



**Selection**, to select specific regions of the image



**Move**, to shift selected areas



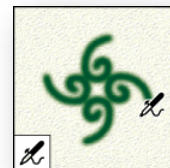
**Lasso**, for freehand selections



**Magic wand**, for the selection of uniform areas



**Crop**, to create image slices

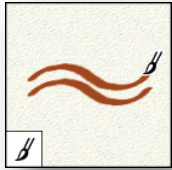


**Airbrush**, to draw with “soft” edges

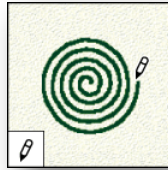


## ... local adjustments...

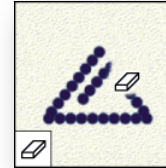
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**Brush**, to create brushworks



**Pencil**, to draw with sharp edges



**Eraser**, to remove parts of an image



**Paint bucket**, to color uniform areas



**Clone stamp**, to replicate parts of an image



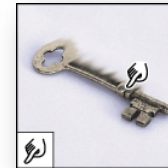
**Text**, to write text



**Blur**, to smooth sharp edges



**Sharpen**, to make edges more definite



**Smudge**, to smear colors

## ... local adjustments

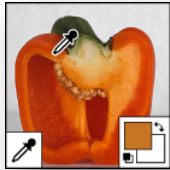
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**Dodge**, to  
lighten parts of  
an image



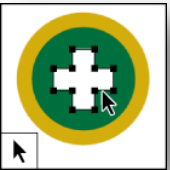
**Pen**, to draw paths



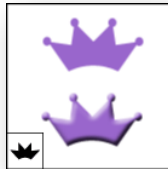
**Color picker**,  
to select a color  
from an image



**Measure**, to  
measure  
distances,  
angles, ...



**Path selection**,  
to select paths



**Rectangle, ellipse,  
polygon, line,  
custom shape**, to  
draw shapes

Each tool has its  
own parameters  
Example: *Paintbrush*  
in GIMP



# The importance of *layers*

Layers allow an image to be modified “gradually”

Similar to drawing on overlapped transparencies... changes in one layer do not affect the content of the others

